

Generation Interconnection System Impact Study Report

For Queue Project Q22-2031

Pinellas County Resource Recovery 100 kV

May 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2031. The Interconnection Customer proposes a 135.1 MW battery energy storage facility interconnecting at the Pinellas County Resource Recovery 100 kV substation in the Ohio area. The request was received on March 11, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 649 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,613,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PINELLAS COUNTY RESOURCE RECOVERY / MARANA 100 KV	IOLA / WAUSAU PAPER MIDDLETOWN 100 KV line	153.0	131	56.5
MARANA / REENERGY STRATTON LLC 100 KV	IOLA / WAUSAU PAPER MIDDLETOWN 100 KV	147.4	148	56.5
CONNERSVILLE / RICHARD M FLYNN 100 KV	WOODSDALE / OROSI 230 KV	127.5	141	18.1
WAUSAU PAPER MIDDLETOWN SUBSTATION / IOLA 100 KV line	MARANA / REENERGY STRATTON LLC 100 KV	123.1	172	49.5
SEMINOLA / WAUSAU PAPER MIDDLETOWN 100 KV	REENERGY STRATTON LLC / MARANA SUBSTATION 100 KV	120.0	133	49.5
SEMNOLA / SERPENT 100 KV	REENERGY STRATTON LLC / MARANA 100 KV	110.1	124	49.5
BLANEY PARK SW STA / OTTER VLY 100 KV Line 1	OTTER VLY SUBSTATION / BLAEY PARK SW STA 100 KV	107.6	151	5.8
BLANEY PARK SW STA / OTTER VALLEY 100 KV #2	BLANEY PARK SW STA SUBSTATION / OTTE VLY 100 KV Cir. 1	107.6	151	5.8
WARNER / IOLA 100 KV	MIDDLETOWN COKE COMPANY LLC / OROSI 100 KV	107.1	170	26.5
MIDDLETOWN COKE COMPANY LLC / OROSI 100 KV L	WARNER / OROSI 100 KV line	104.0	213	27.4
WARNER SUBSTATION / OROSI 100 KV	OROSI / MIDDLETOWN COKE COMPANY LLC 100 KV	103.2	207	26.5
SERPENT / PINELLAS COUNTY RESOURCE RECOVERY 100 KV	REENERGY STRATTON LLC / MARANA 100 KV	102.3	111	49.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PINELLAS COUNTY RESOURCE RECOVERY / MARANA 100 KV	Rebuild line with higher-capacity conductor	\$1,269,000	100% (\$1,269,000)
MARANA / REENERGY STRATTON LLC 100 KV	Rebuild line with higher-capacity conductor	\$10,031,000	100% (\$10,031,000)
RICHARD M FLYNN / CONNERSVILLE SUBSTATION 100 KV	Rebuild line with higher-capacity conductor	\$3,166,000	100% (\$3,166,000)
WAUSAU PAPER MIDDLETOWN SUBSTATION / IOLA 100 KV line	Rebuild line with higher-capacity conductor	\$2,905,000	100% (\$2,905,000)
SEMINOLA / WAUSAU PAPER MIDDLETOWN 100 KV	Rebuild line with higher-capacity conductor	\$6,063,000	100% (\$6,063,000)
SEMINOLA SUBSTATION / SERPENT 100 KV	Rebuild line with higher-capacity conductor	\$2,109,000	100% (\$2,109,000)
BLANEY PARK SW STA / OTTER VLY 100 KV Line 1	Reconductor line	\$321,000	37% (\$120,000)
OTTE VLY SUBSTATION / BLANEY PARK SW STA 100 KV Ckt 2	Reconductor line	\$471,000	37% (\$176,000)
IOLA / WARNER SUBSTATION 100 KV	Reconductor line	\$172,000	100% (\$172,000)
OROSI / MIDDLETOWN COKE COMPANY LLC 100 KV	Reconductor line	\$237,000	100% (\$237,000)
WARNER SUBSTATION / OROSI 100 KV	Reconductor line	\$805,000	100% (\$805,000)
PINELLAS COUNTY RESOURCE RECOVERY / SERPENT 100 KV	Reconductor line	\$560,000	100% (\$560,000)
	Total allocated to project		\$27,613,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.